

Preserve the capture first. Optimize delivery second.

The ten references below solve different problems: bulk video streaming, commercial catalogue QA, manifest-led commercial distribution, multimodal loading, native asynchronous capture, ground-truth evaluation, reproducible export, human/robot normalization, cross-dataset robot training, and human-to-robot action transfer. Actuate needs a verified source package before any training representation.

01 / RAW

Native Panoculon Trinet files

MP4, IMU, VTS, optional TEL, JSON, and embedded metadata remain byte-identical.

02 / DECODE

Native-rate IMU

Confirmed IMU fields and original timestamps become one typed table without interpolation.

03 / VERIFY

Traceable QC

Measured facts, deterministic diagnoses, transformations, and human decisions stay distinct.

04 / DELIVER

Inspectable episodes

One capture folder can be opened by tools, the CLI, or the shared review service.

Chosen v1 package

```
dataset/
├── README.md
├── episodes.csv
├── calibration/                # when supplied and bound
│   └── calibration_000001.json
├── episodes/episode_000001/
│   ├── meta.json
│   ├── raw/                   # byte-identical inputs
│   └── derived/
│       ├── imu.parquet
│       ├── frame_timing.parquet
│       └── telemetry.parquet   # when present
└── previews/                  # when supplied
```

Meaning of each layer

README.md	Human guide to fields, limitations, inspection, and loading.
episodes.csv	One factual row per delivered episode with stable identity and source provenance.
calibration/	Machine-readable calibration only when explicitly bound to the recorded rig.
raw/	Authoritative source bytes. Hashes establish identity, not quality.
derived/	Native-rate IMU and per-frame timing tables, plus telemetry when present.
previews/	Clearly separated supplied visualizations. Internal QC verdicts are not delivered.

Evidence boundary: richer working indexes and review caches do not change this customer-facing v1 package.

How access pattern shapes the package

The middle column states the published format. The next explains its documented fit using scale and loader behavior. The last combines the useful lesson and the boundary for Actuate.

Company / reference	Exact published format and access path	Why this choice fits that product	Combined lesson and Actuate decision
Public dataset card Build AI Egocentric-10K	factory_N/workers/worker_N/ contains one intrinsics.json plus WebDataset TAR shards capped at 1 GB. Every shard contains same-stem H.265 .mp4 + .json samples. JSON records factory, worker, clip index, duration, width, height, FPS, bytes, and codec. Hugging Face examples stream all shards or select factories/workers.	The card documents 192,900 clips and about 18 TB. Factory/worker paths enable subset loading; bounded TARs enable streaming without expanding millions of loose files; intrinsics are stored once per worker/device.	Adopt: stable media/metadata stems, measured media facts, device calibration, and optional sharding for transfer. Do not copy as canonical: TARs hinder recurring file-level inspection/replacement, and the video-only format has no native IMU or hardware-clock contract.
Publicly documented, files gated Human Archive 500h / HA-Ego-Samples	commercial/factory, commercial/hospitality, and residential contain personN/. Each has personN_segments.json and H.264 personN_segmentK.mp4. JSON stores person totals plus each segment's task, environment, duration, dimensions, FPS, bytes, and codec.	The product is a 10,633-segment commercial video catalogue. Its card says every MP4 is fully ffprompt-verified; JSON/media pairing, indices, durations, corrupt/short clips, and whole-corpus duplicates are checked. One compact JSON exposes the person's catalogue.	Adopt: playable MP4, compact inventory, strict pairing, measured properties, full-file validation, and duplicate detection. Do not copy: task/environment/person remain unknown for Actuate unless supplied; there is no sensor-clock representation. The card says "LeRobot v3," but the shown tree is custom MP4 + JSON.
Publicly documented, files gated Human Archive HA-Multi-Samples-v2	episodes/ep_N/ contains four view MP4s, rectified stereo MP4s, tactile (N,256) NumPy arrays, glove IMU (N,12), body/head IMU arrays, and meta.json. Global meta/ holds episodes.csv, tasks.csv, calibration.json, and stats.json. Sensor rows align to 30 FPS frames.	The card says v2 was reorganized into episode/modality folders for easier browsing and partial loading without a LeRobot dependency. Separate arrays make modality selection direct; CSVs index the corpus; shared calibration and fixed shapes simplify frame-aligned access.	Adopt: self-contained episode folders, separated modalities, a global catalogue, calibration, and statistics. Do not copy as canonical: frame alignment is derived convenience. Native Panoculon Trinet IMU/timestamps must survive first. The card conflicts on current depth availability, so this comparison makes no depth claim.
Official gated commercial release Lightwheel EgoSuite-Open100K	The collection separates EgoStandard (90,000 planned head-view hours), EgoPro (10,000 planned head-and-wrist hours), and a 50-hour EgoDemo. EgoDemo supplies raw MP4, LeRobot v3, MCAP, and manifests. One source_episode_uuid joins alternative formats; manifests carry stable IDs, provenance, task/scene, duration, paths, modality flags, format, and SHA-256.	Lightwheel sells controlled-access data while exposing a developer sample and common contract. Raw and annotated SKUs stay distinct, formats are not double-counted, and manifests are the source of truth for selective access to mutable buckets. The public contract documents no calibration package.	Adopt: neutral dataset/version naming, stable episode provenance, manifest-led selection, raw/annotated separation, checksums, and one identity across buyer formats. Do not copy blindly: task/pose fields need source evidence; adapters stay optional; mutable storage does not replace an immutable customer dataset.

Human Archive correction: these are two distinct public schemas. The 500-hour MP4 + per-person JSON product is not the multimodal v2 episode package and is not the separate six-camera Human Archive datasets.

Capture fidelity and derived evidence

These references separate native sensor recording from derived research outputs. That separation is the decisive pattern for Actuate.

Company / reference	Exact published format and access path	Why this choice fits that product	Combined lesson and Actuate decision
Official format + SDK Meta Project Aria VRS	One .vrs stores typed per-sensor Configuration, State, and Data records for camera, audio, IMU, magnetometer, and other streams. Configuration blobs carry static sensor information; calibration is addressed by stream label. Same-device sensors share DEVICE_TIME; APIs query by timestamp and declared time domain. MPS outputs stay separate.	Meta documents VRS for multimodal XR recording, large files, timestamp-order playback, and random access. It preserves asynchronous native rates while an integrated device supplies stream IDs, calibration, one capture clock, integrity tooling, and timestamp-query APIs.	Adopt: explicit clock domains, native-rate streams, capture-versus-arrival semantics, configuration/calibration provenance, integrity checks, and timestamp queries. Do not convert Panoculon Trinet to VRS: repacking cannot recreate Aria hardware guarantees and would promote a derived container above original sidecars.
Official dataset + toolkit Meta HOT3D Full sequences + HOT3D-Clips	Full folders include recording.vrs, metadata, object definitions, headset trajectory, MANO/UmeTrack trajectories, boxes, MPS outputs, and timestamp/stream mask CSVs. Separately, 3,832 five-second WebDataset clips contain 150 keyed frames, three camera JPEGs per frame, camera/hand/hand-crop/object/info JSON, and shared hand-shape data.	The full VRS-based dataset supports sequence research; fixed WebDataset clips serve BOP and hand-tracking evaluation. Independent masks encode pose availability, visibility, exposure, and manual-QA pass, so consumers compose a validity rule without deleting annotations or confusing availability with accuracy.	Adopt: property-specific validity, timestamp-level evidence, separate manual decisions, and a hard boundary between full recordings and derived clips. Do not copy: hand/object ground truth and benchmark packaging are future enrichment or adapters, not Actuate v1 source evidence.
Official dataset + open SDK FPV Labs Stera-10M + Stera SDK	The iPhone app records RGB, depth, pose/intrinsics, IMU, tracking, transforms, map, and trajectory as ROS2 topics in raw MCAP. Published sessions contain rgb.mp4, annotation.hdf5, hierarchy.json, mesh.ply, thumbnail, six calibration files, and full/preview Rerun recordings. HDF5 separates depth, pose, native-rate IMU, hand pose, and metadata; aligned arrays include validity.	The documented loop targets reproducible hour-plus multimodal collection on commodity iPhone Pro hardware. MCAP serves capture; HDF5 supports selective numeric reads; MP4 remains playable; PLY carries geometry; calibration is explicit; Rerun supplies review; the Apache-licensed SDK reproduces processing.	Adopt: raw/processed separation, natural-rate IMU, explicit validity/units/frames, model provenance, self-contained sessions, and review artifacts. Do not copy as canonical: Actuate's current inputs lack Stera's derived modalities. Stera docs and current exporter also conflict on a depth timestamp and hand coordinate framing, proving nominal schemas still require file validation.

Shared pattern: Aria preserves integrated-device capture; HOT3D layers validity and benchmarks on top; Stera publishes a capture-to-export path. None justifies replacing uncertain Panoculon Trinet originals with a more polished derived container.

When normalization becomes useful

These rows belong later in the pipeline. They require calibrated motion, robot trajectory semantics, or a declared human-to-robot action representation.

Company / reference	Exact published format or representation	Why this choice fits that product	Combined lesson and Actuate decision
Official code + converters Georgia Tech EgoVerse / EgoMimic	Current storage is one <episode_hash>.zarr/ v3 group, a same-hash MP4 preview, and a PostgreSQL catalogue row. Frame-indexed arrays include JPEG-byte images.front_1, head/wrist/EE poses, optional MANO points, RGB timestamp values, and annotations; root metadata carries embodiment, total_frames, and mandatory 3x4 intrinsics. The Aria converter combines VRS with MPS hand/SLAM CSVs and removes invalid-pose frames.	The repository co-trains on human and robot demonstrations. One chunked episode interface and catalogue batch images, poses, actions, annotations, and overlays; embodiment metadata selects the representation. total_frames bounds padded tails and absent modalities are normally omitted.	Adopt later: explicit source/embodiment, geometry-gated calibration, authoritative length, versioned conversion, and catalogue-backed views. Do not copy for v1: it needs SLAM/hand actions, aligns to video frames, omits high-rate IMU, and drops invalid frames. Current guide/code also disagree on timestamp domain and calibration provenance.
Official unified schema Google DeepMind Open X-Embodiment / RLDS	Open X distributes versioned TFDS directories, typically sharded TFRecord files plus dataset/feature metadata, loaded from GCS or local storage. RLDS is the logical read-time schema, not one container: episode dictionaries contain metadata and nested ordered steps. Steps require is_first/is_last; observation, action, reward, terminal, discount, fields, and shapes remain dataset-specific.	Each native robot dataset uses its own TFDS builder and feature mapping, then exposes one episode/step interface for downstream mixtures. Training can interleave heterogeneous embodiments while retaining contributor-specific observations/actions. RLDS does not impose one universal action vector.	Adopt only as a buyer adapter: useful after genuine episode, observation, action, state, task, and terminal semantics exist. Do not use as canonical: Actuate's current captures lack robot actions/rewards/termination. Inventing fields or resampling IMU merely to make steps creates false semantics and hides native evidence.
Papered representation, no public package NVIDIA GEAR EgoScale	No public on-disk dataset schema or loader is released; the official page says GitHub "Coming Soon." The paper defines 30 FPS RGB plus estimated camera SE(3) and 21 hand-keypoint transforms, converted to relative wrist motion and optimization-retargeted 22-DoF Sharpa targets. Precision/alignment stages add Vision Pro, Vive wrist pose, Manus 25-joint transforms, and synchronized views.	EgoScale is a cross-embodiment training recipe, not neutral delivery. Relative wrist motion removes global camera movement; retargeted joints provide finger supervision; small embodiment adapters and aligned human/robot data connect that shared representation to executable control.	Future lesson: pair raw scale with precision anchors; declare coordinates, target embodiment, validity, provenance, and evaluation for wrist/finger actions. Do not call it a format: public storage fields are undocumented, transformations are model-specific, and the paper describes noisy off-the-shelf estimates.

Chosen source contract

Byte-identical native files + manifest CSV + native-rate IMU Parquet + one QC JSON.

Why it fits Actuate

The product verifies uncertain sidecars and clocks before training semantics exist.

Adapters, not authority

Zarr, RLDS, LeRobot, MCAP, HDF5, and action targets follow only from measured buyer needs.